MOBILE DRILL BRAIN

Calibration Technician Training Module 005

LP, LE, Depth Intervals, and Field Measurement Accuracy

Version v0.1 - Review Deliverable

Document field	Value
Document type	Workspace 006-B technician training module
Module	005 - LP, LE, Depth Intervals, and Field Measurement Accuracy
Version	v0.1 Review Deliverable
Prepared	July 2026
Source basis	SPT Introduction Draft v6; Chapter 2 Draft and Field Lessons; Calibration Data Collection Cheat Sheet; Module 004 SME-approved workflow clarifications
Training lane	Calibration Technician Training - measurement accuracy and report usability
Status	Internal training draft; not a final field SOP or analyzer manufacturer manual

Locked module purpose

Teach a calibration technician how to correctly understand, measure, enter, and document the field values that make calibration data usable for report building.

Module 004 taught the software workflow. Module 005 focuses on the values that must be correct inside that workflow: depth interval, sample interval, LP, LE, rod type, 6-inch blow-count increments, refusal documentation, and field notes.

Core training rule

Good data collection is not just getting the analyzer to record hammer blows.

Good data collection means the saved data file, handwritten field record, depth interval, LP, LE, and field conditions all agree.

1. Source basis and training boundary

This module uses the SPT introduction material for the 6-inch increment and N-value foundation, the Chapter 2 field lessons for rod length/refusal/data-quality context, and the calibration data collection cheat sheet for the analyzer fields that must be entered for each data set.

Training boundary

This module does not teach final engineering interpretation. The technician is not deciding soil capacity, foundation design, liquefaction risk, or final N60 interpretation. The technician collects accurate, reviewable field data so the report builder and engineering/customer side have reliable information to work from.

2. Learning outcomes

1. Define LP correctly as length of penetration.
2. Define LE as the measured distance from the sampler soil contact point - the bottom of the hole - to the first sensor location on the instrumented rod.
3. Explain why LP and LE cannot be guessed or copied from the previous interval.
4. Enter the correct depth interval in feet.
5. Explain the difference between an 18-inch and 24-inch sample interval.
6. Record blow counts on paper for each 6-inch increment.
7. Explain why the software blow total is not enough by itself.
8. Recognize refusal when the sampler does not advance a full 6 inches in 50 blows.
9. Explain how mixed depths and mixed intervals can still support a cohesive report if documented correctly.

10. Identify when missing or wrong field values make a data set questionable.

3. Why measurement accuracy matters

The analyzer records energy data, but the report depends on more than the graph. A report builder needs to know exactly what the analyzer data represents. If the file says one thing and the field notes say another, the report builder may not be able to use the data confidently.

Bad measurement documentation can create problems such as:

- data file tied to the wrong depth
- LP/LE mismatch
- unclear sample interval
- missing 6-inch blow counts
- refusal not documented
- questionable data treated as clean
- report builder unable to explain why one interval looks different from another

Training rule

The technician is responsible for making the field record match the saved analyzer file.

4. Depth interval

The depth interval identifies where the sample starts and how far the sampler travels into the soil. It is entered in feet.

Example	Meaning
20.0 to 21.5 ft	18-inch sample interval
20.0 to 22.0 ft	24-inch sample interval

Training rule

The depth interval must match the actual sampled interval.

Do not enter a 20.0-22.0 ft interval if the sampler only traveled from 20.0 to 21.5 ft.

5. Sample interval

The sample interval is the amount of soil penetration being collected for that run.

Interval	Feet	Training note
18 inches	1.5 ft	Minimum sampler interval for proper data collection
24 inches	2.0 ft	Commonly preferred when available because it collects more data and makes back-end report math easier

The technician will see both 18-inch and 24-inch intervals in the field.

Training rule

Record what actually happened, not what would have been convenient.

6. Production drilling caveat

On production drilling jobs, calibration should control data quality as much as possible, but calibration does not override the sampling standards, intervals, or frequencies set by the entity paying for the drilling.

Different projects may require different sample intervals and sampling frequencies. The calibration technician must work within those requirements.

If the production sampling requirement is met but insufficient calibration data has been collected, data collection can return to the standard calibration approach afterward to gather enough usable data.

Training rule

Different depths and different sample intervals can still create a cohesive calibration report if everything is recorded correctly.

Required record for mixed intervals:

- depth interval
- sample interval
- LP
- LE
- blow counts by 6-inch increment
- field notes
- refusal or unusual advancement

7. LP - Length of Penetration

LP means length of penetration.

LP definition

For Mobile Drill calibration training, LP is the distance from the top of soil / ground surface to the bottom of the hole.

LP should represent the actual hole-depth context for the data being collected. Do not copy the previous LP value unless the current interval actually uses the same value.

One numeric example

If the sample starts at 20.0 ft and advances to 21.5 ft, the depth interval is 20.0-21.5 ft. In this normal setup, LP is tied to the 20.0 ft bottom-of-hole context at the start of the run. The exact entered value must match the actual field setup.

8. LE - sampler soil contact point to first sensor

LE definition

LE means the measured distance from the sampler soil contact point - the bottom of the hole - to the first sensor on the instrumented rod.

LE must be measured. A common value may be around 3.5 ft because of how rod lengths and the instrumented rod setup usually work out, but the value must still be confirmed.

Training rule

LE is not a memory value. LE is a measurement.

Do not enter 3.5 ft just because it is common.

9. Why LP and LE change by interval

Each data set needs its own correct depth and rod-length context. The calibration data collection cheat sheet notes that LP and LE should be different for each data set collected and that changing LP and LE creates the new file context for stored data.

Training rule

Each interval gets its own LP, LE, and depth interval context.

If the technician drills deeper, the values must be updated.

10. Rod type and rod condition

Rod type must be confirmed for the job. The technician may be using AWJ or NWJ instrumented rods depending on the rig and job setup. Rod condition and rod connections also matter.

Chapter 2 field lessons emphasize that rod joints must be clean, tight, and flush because small gaps reduce energy transfer.

Check item	Why it matters
Correct instrumented rod type	AWJ/NWJ mismatch creates wrong setup context
Rod joints clean, tight, and flush	Small gaps can reduce energy transfer
Anvil threaded flush into instrumented rod	Bad fit can distort readings or damage equipment
No obvious rod damage	Damaged rods can affect the response of the rod string
Instrumented rod protected from HSA/borehole	Prevents sensor damage

11. Blow counts by 6-inch increment

The software records total blow count. The paper record must capture each 6-inch increment. This matters because SPT sampling and N-value logic are built around 6-inch increments.

The paper record helps the report builder understand:

- how the sampler advanced
- whether one 6-inch section behaved differently
- whether refusal was developing
- whether the sample interval was 18 inches or 24 inches
- whether data should be questioned or explained

18-inch interval	24-inch interval
0-6 in. _____ blows	0-6 in. _____ blows
6-12 in. _____ blows	6-12 in. _____ blows
12-18 in. _____ blows	12-18 in. _____ blows
	18-24 in. _____ blows

Training rule

The software total does not replace the paper 6-inch breakdown.

12. Refusal

Concrete refusal rule

For training purposes, refusal means the sampler does not advance a full 6 inches in 50 blows. This is an industry-standard threshold that drillers operate by.

Calibration data collected during refusal should not be treated as normal clean calibration data.

If refusal occurs:

- stop treating that interval as normal clean calibration data
- record the blow count
- record the depth
- record how far the sampler advanced
- document the field condition
- follow the proper field decision or escalation path

Training rule

Refusal is not just a hard drilling problem. It is a data-quality issue.

13. Loose soil, rapid advancement, and instrumented rod protection

Data quality problems do not only come from refusal. The opposite condition can also matter: soil may be loose enough that the sampler advances very quickly.

CRITICAL WARNING - protect the instrumented rod

Do not let the instrumented rod enter the hollow stem auger or borehole.

The technician should observe the first 6 blows to make sure the soil is not so loose that the rod and sampler are advancing too quickly.

Before collection begins, the operator should be told to stop hammer operation immediately if the instrumented rod is in danger of entering the hollow stem auger or borehole.

If rapid advancement occurs:

- tell the operator to stop hammer operation if the instrumented rod is at risk
- protect the sensors
- document the condition
- correct setup or move to a safer collection approach

This is both a data-quality issue and an equipment-protection issue.

14. Field notes

The technician should document anything that explains the data. Good field notes allow the report builder to understand what happened without standing next to the tech in the field.

Useful field note topic	Example
Refusal or rapid advancement	50 blows without 6 inches of movement; sudden 6-inch sink in one blow
Ground conditions	loose soil, shallow bedrock, cobbles/gravel, unstable borehole
Hammer behavior	RPM concern, hydraulic pressure concern, hammer cycling issue, suspected short stroke
Rod/sensor issues	rod joint issue, sensor issue, rerun, changed interval
Production constraints	project-required interval, customer/driller pressure to move faster

Training rule

If the report builder was not standing next to you, would your notes explain what happened?

15. Mixed intervals and cohesive reports

A report can still be cohesive when data comes from different depths or intervals. The key is documentation.

Mixed data may still be usable when the technician correctly records:

- where each interval started
- where each interval ended
- whether the interval was 18 inches or 24 inches
- LP for each interval
- LE for each interval
- 6-inch blow-count breakdown
- reason for any interval change
- refusal, rapid advancement, or abnormal condition

Training rule

Mixed intervals are not automatically bad. Undocumented intervals are bad.

16. When a data set becomes questionable

A data set may become questionable when:

- depth interval is wrong or missing
- LP is wrong or copied from the previous run
- LE was guessed
- 6-inch blow counts were not recorded
- refusal occurred but was not documented
- rapid advancement occurred but was not documented
- instrumented rod entered or nearly entered the HSA/borehole
- sensor issue occurred during collection
- rod/anvil connection was not flush
- production sampling requirement changed the interval but notes do not explain it

Training rule

Questionable does not always mean unusable, but it must be documented and reviewed.

17. Common mistakes

Mistake	Why it matters	Correct behavior
Guessing LE	Wrong sensor distance context	Measure LE
Copying LP/LE from prior run	New interval may have different depth context	Confirm each interval
Entering 20.0-22.0 ft when sample was 20.0-21.5 ft	File does not match field reality	Enter actual interval
Relying only on software blow total	Report needs 6-inch increments	Record paper blow counts
Failing to document refusal	Report builder may treat bad data as normal	Record 50 blows / less than 6 inches advancement
Ignoring rapid advancement	Instrumented rod/sensors may be damaged	Watch first 6 blows and stop if needed
Not noting production interval requirements	Mixed intervals may look unexplained	Document project-driven interval changes
Treating mixed intervals as unusable	Good data may be rejected unnecessarily	Use documentation to support report context

18. Training scenario

A production drilling project requires 18-inch samples at the calibration depths. The tech prefers 24-inch samples because they make reporting easier, but the project sampling requirement is already set.

At the first interval, the tech enters 20.0-22.0 ft out of habit. The driller only advances the sampler to 21.5 ft. The software records total blows, but the tech forgets to write the 6-inch blow-count breakdown.

Correct trainee response should include:

- Calibration does not override the project sampling requirement.
- The depth interval should be corrected to 20.0-21.5 ft.
- LP and LE should match the actual interval/depth context.
- The 6-inch blow-count breakdown is required for report building.
- If the mistake is only discovered later, document it clearly so the report builder can address it during report building.
- If the mistake is caught on site, best practice is to correct it in the field by drilling deeper and collecting another data set with proper depth interval, LP, LE, and 6-inch blow-count documentation.
- Future intervals must be recorded accurately even if they are mixed 18-inch and 24-inch intervals.

19. Knowledge check

#	Question	Answer key
1	What does LP stand for?	Length of penetration.
2	What is LP for this training module?	Distance from top of soil / ground surface to bottom of hole.
3	Should LE be guessed if it is usually around 3.5 ft?	No. LE must be measured.
4	What does the depth interval field represent?	Where the sample starts and how far the sampler travels into soil, entered in feet.
5	What is the minimum sampler interval for proper data collection?	18 inches / 1.5 ft.
6	Why is a 24-inch interval often preferred when available?	It collects more data and makes back-end report math easier.
7	Can different depths and intervals be mixed in one report?	Yes, if depth interval, sample interval, LP, LE, 6-inch blow counts, and notes are correctly recorded.
8	Why are paper blow counts needed if software records total blows?	The report needs the individual 6-inch breakdown and refusal awareness.
9	What is the refusal rule?	Sampler does not move a full 6 inches in 50 blows.
10	Why should the tech observe the first 6 blows?	To make sure the sampler/rod is not advancing too quickly and risking the instrumented rod entering the HSA/borehole.
11	What should happen if production sampling requirements conflict with preferred calibration intervals?	Work within production requirements and document accurately.
12	What is worse: mixed intervals or undocumented intervals?	Undocumented intervals.

20. Practical signoff checklist

Signoff item	Pass standard	Trainer initials/date
Defines LP	Trainee defines LP as length of penetration and explains field meaning	
Measures/understands LE	Trainee states LE must be measured and cannot be guessed	
Enters depth interval	Trainee correctly enters 18-inch and 24-inch intervals in feet	
Handles production caveat	Trainee explains that calibration works within project sampling requirements	
Records blow counts	Trainee records every 6-inch increment on paper	
Identifies refusal	Trainee identifies refusal as less than 6 inches in 50 blows	
Watches rapid advancement	Trainee explains first-6-blow observation and instrumented rod protection	
Documents mixed intervals	Trainee shows how mixed depths/intervals can still support a report	
Recognizes questionable data	Trainee identifies when missing values or bad notes require review	

21. Next module inputs

Module 006 should cover data quality, refusal handling, bad data response, sensor issues, field troubleshooting, reruns, and escalation. Module 005 intentionally stops at accurate field measurement and documentation, because deeper troubleshooting belongs in the next training module.

Revision history

Version	Status	Change
v0.1	Review deliverable	Initial Module 005 produced from approved in-thread draft and SME corrections on LP, LE, sampler intervals, production caveat, instrumented rod protection, and report-building documentation.